Exp No: 6 VERIFICATION OF MAXIMUM POWER TRANSFER THEOREM

Pg. No

Date: 25/2/26

AIM:

To measure the power absorbed in a load and to verify that the power absorbed in a load is maximum only when load resistance is equal to the source resistance.

APPARATUS REQUIRED:

S.NO	Name of the apparatus	Range / Rating	Quantity
1	Voltmeter	(0-15V) MC	1
2	Ammeter	(0-500mA) MC	1
3	Resistor	560 Ω, 470 Ω	2, 1
4	RSP (DC)	15 V	1

PROCEDURE:

1) Make connection as per the circuit diagram.

2) Change the resistors R_L whose value close to R_{th}, measure the corresponding V_L, I_L and calculate the power P_L and enter into the table (2).

3) Plot a graph between R_L and P_L and find the R_L corresponding to maximum power transfer.

4) Verify the measured values of R_L at max. power transfer as same as calculated and found graphically.

Mark split-up.

SNo	Description	Marks alloted	Marks obtained.
1	Experimental setup/ Connections	20	20
2	Observation / Reading	20	20
3	Calculation / Results	20	20
4	Viva	20	20
5	Record	20	20
		Total	100

100

17/3/26

RESULT:

Thus maximum power transfer theorem is verified. Practically and theoretically..

MODEL CALCULATION.

$$V_{Th} = \frac{V \times R_2}{R_1 + R_2} = \frac{10 \times 560}{500} = 5V$$

R_{Th} = R_1 and R_2 is parallel.

$$R_{Th} = \frac{560 \times 560}{2 \times 560} + 470$$

$$= 280 + 470$$

$$= 750\Omega.$$

$R_L = R_{Th}$.

$$P_L = \frac{(V_{Th})^2}{4 \times R_{Th}} = \frac{5^2}{4 \times 750}$$

$$= \frac{25}{3000} = 0.0083 \text{ mW}$$

$P_L = 0.0085$ mW.

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{5}{1500} = 0.0033.$$

$I_L = 3.33$ mA.

$$V_L = I_L \times R_L = 3.33 \times 10^{-3} \times 750$$

$V_L = 2.5V$.

TABULATION:

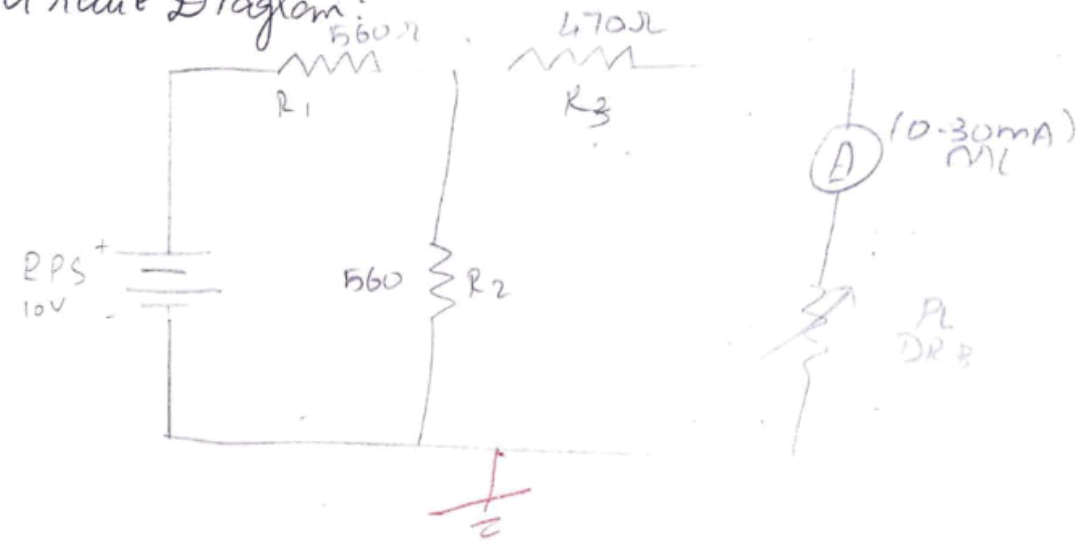
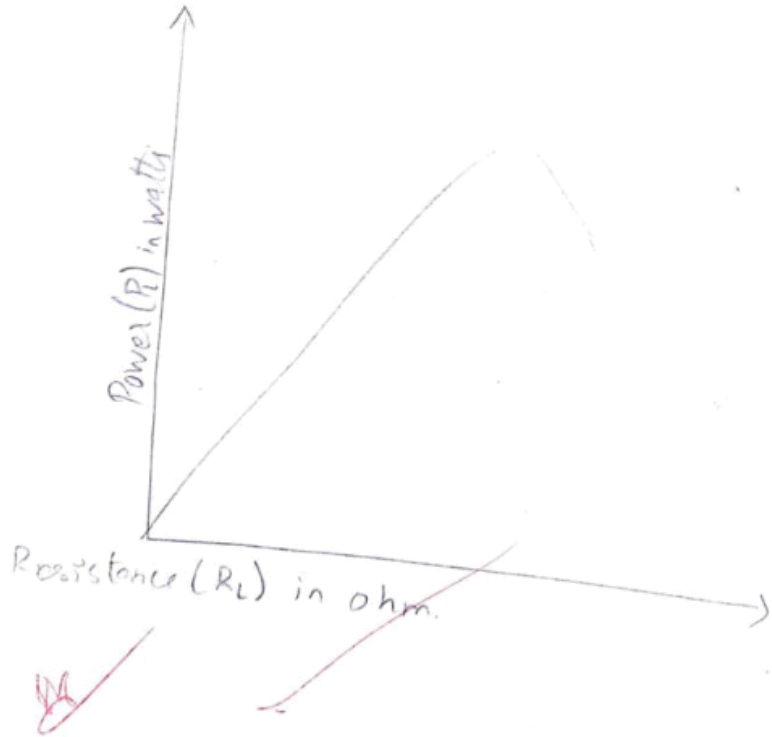
TABLE(1) : For theoretical Calculation : $R_{Th} = 750\Omega$

S.No	Load Resistance (R_L) in Ohm	Load current (I_L) in mA	Load Voltage (V_2) in volts	Load Power (P_L) in watts
1	750	3.33	2.5	8.33×10^{-3}
2	470	4.09	1.92	7.85×10^{-3}
3	560	3.81	2.13	8.115×10^{-3}
4	800	3.12	2.65	8.26×10^{-3}
5	1KΩ	2.85	2.85	8.12×10^{-3}

TABLE(2) : For Practical Calculation:

SNo	Load Resistance (R_L) in Ohm	Load current (I_L) in amps	Load Voltage (V_L) in Volts	Load Power (P_L) in watts.
1	470	4.09	1.92	7.85×10^{-3}
2	560	3.81	2.13	8.115×10^{-3}
3	750	3.33	2.5	8.3×10^{-3}
4	800	3.12	2.65	8.26×10^{-3}
5	1KΩ	2.85	2.85	8.12×10^{-3}

Circuit Diagram:



MODEL GRAPH